

AI-Powered Dashboards for Cohort Study Monitoring and Analysis

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Abstract

We present an AI-powered dashboard system designed to improve the lifecycle of cohort studies by offering intelligent support during both data collection and post-study analysis. The dashboard combines natural language interfaces with generative agents to assist in data monitoring, chart creation, anomaly detection, and result interpretation. This paper outlines a modular architecture for flexible and scalable deployment and demonstrates its potential through early-stage prototypes developed for medical research.

CCS Concepts

• *Applied computing* → *Life and medical sciences*; • *Computing methodologies* → *Artificial intelligence*;

1. Introduction

Researchers managing large-scale cohort studies often face operational bottlenecks, such as delays in identifying missing assessments or challenges in comparing recruitment across multiple sites. Addressing these issues typically requires support from technical staff, given the complexity of structured database queries [JNQ11].

To reduce these barriers, we are developing *CohortCanvas*—an AI-augmented dashboard that supports both real-time monitoring during data acquisition and flexible post-study analysis. It enables users to pose high-level questions (e.g., “Show recruitment trends over the last 3 months by site”) and receive visual results without writing any code.

Key features:

- **Monitoring phase:** Real-time participant tracking, protocol adherence, and detection of missing or anomalous data [HWS*24].
- **Analysis phase:** Natural-language-driven filtering, subgroup comparison, and automated chart generation.

CohortCanvas leverages large language models (LLMs), retrieval-augmented generation (RAG), and semantic embeddings to lower the technical threshold for working with cohort data. These techniques translate user queries into structured operations over cohort datasets [XZZZ25], improving accessibility across disciplines and supporting more inclusive, real-time data exploration.

2. Methods

CohortCanvas is designed as a modular system composed of independent agents, each responsible for a specific aspect of user interaction, data access, or visualization. The system aims to reduce the technical complexity of interacting with cohort data while preserving flexibility and security.

Orchestration and Agent Architecture: At the core of the platform is an orchestration layer that receives natural language requests from the user and determines the appropriate agent to handle the task. For instance, a request to visualize a metric over time is routed to the chart generation agent, while a request to find variable names is handled by the variable finder agent.

Semantic Variable Finder: The variable finder agent maps natural language phrases to dataset variables using a semantic embedding pipeline. User queries and variable metadata (from the data dictionary) are encoded using the *sentence-transformers* library and stored in a PostgreSQL database with PGVector support. During a query, the system retrieves the top matches based on cosine similarity. This allows flexible queries such as “anxiety score” to identify variables like `GAD7_total` — the Generalized Anxiety Disorder (GAD-7) scale, even if the wording differs.

SQL and Chart Generator: Once the relevant variables are identified, a chart generation agent uses a locally hosted LLM (LLaMA3 via Ollama) to generate SQL queries and Vega-Lite specifications. Prompts are tightly structured to reduce hallucination and ensure syntactic validity. This enables automatic chart generation for user questions like “Compare recruitment trends by site over the last three months.”

Interactive Dashboard Interface: The user interface is developed in Next.js and uses the React Grid Layout library to support a modular, section-based design. Users can create, resize, and arrange individual chart blocks on the canvas. Each block can independently connect to the AI assistant, enabling localized requests such as filtering by age group or comparing outcomes across subgroups. Linked blocks share filters or selections, allowing synchronized updates across charts for comparative analysis.

Backend and Deployment: Data is stored in PostgreSQL and

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